PCA-Derived Factors for Market Regime Prediction

Quant Mentorship Final Project

Arth Tak

November 2025

1. Data and Research Question

Equity sectors often move together, but their correlations evolve across market conditions. This project investigates whether a small number of statistical factors extracted from U.S. sector ETFs using PCA can help predict market regimes, and whether those regimes can improve a simple allocation rule versus buy-and-hold SPY.

I use daily adjusted close prices for SPY and 11 sector ETFs (XLK, XLF, XLE, XLI, XLY, XLP, XLU, XLV, XLB, XLRE, XLC) from 2010–present, downloaded via `yfinance`. After aligning calendars, I compute log returns and standardize them before PCA.

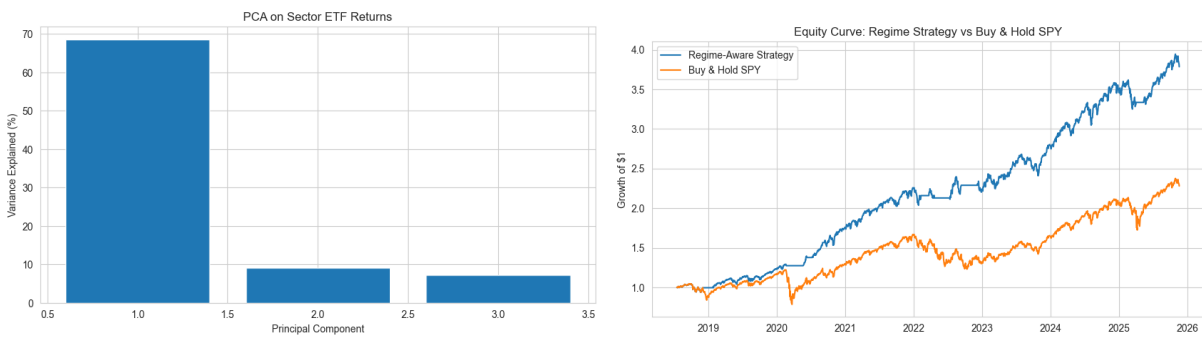

Figure 1: Left: PCA variance explained. Right: Regime-aware strategy vs SPY.

2. Methodology

PCA factors. Let X be the matrix of standardized sector returns. PCA solves

$$\max_{v} \ \mathrm{Var}(Xv)$$

yielding orthogonal components. PC1 captures broad systematic co-movement (a "market factor"), while PC2 typically loads positively on cyclicals and negatively on defensives, representing rotation or risk-on/risk-off structure. PC3 is more idiosyncratic. I retain the first three PCs, which explain the majority of sector variance.

Regimes via Hidden Markov Model (HMM). To infer latent market states, I fit a 3-state Gaussian HMM to daily SPY return and 20-day realized volatility. HMMs model the observation distribution and the transition matrix simultaneously, allowing regimes to evolve smoothly over time. The fitted model naturally produces:

- a low-vol, positive-return state (bull),
- a high-vol, negative-return state (bear),
- a moderate-vol neutral regime.

The smoothing probabilities give the probability of each regime on each day.

Predicting regimes with PCA factors. To evaluate predictability, I construct a binary target corresponding to the high-volatility regime one day ahead:

$$Y_t = \mathbb{1}\{S_{t+1} = \text{high-vol regime}\}$$

and regress this on lagged PCA factors using logistic regression:

$$\mathbb{P}(Y_t = 1 \mid X_t) = \sigma(\beta_1 \mathrm{PC1}_t + \beta_2 \mathrm{PC2}_t + \beta_3 \mathrm{PC3}_t).$$

Training uses the first 70% of the sample; testing uses the final 30%.

Backtest. Using the inferred regime sequence, I implement a simple allocation rule:

$$w_t = \begin{cases} 1.0 & \text{bull regime} \\ 0.5 & \text{neutral regime} \\ 0.0 & \text{bear regime} \end{cases}$$

Daily returns follow $r_t^{strat} = w_t r_t^{SPY}$. I compare to buy-and-hold using annualized return, annualized volatility, Sharpe ratio, and maximum drawdown.

3. Results and Interpretation

PCA structure. The scree plot shows that the first three PCs explain around 84% of total sector-variance. The loading structure confirms strong economic meaning: PC1 behaves like a market beta component; PC2 captures defensive vs. cyclical dispersion; PC3 is small but reflects occasional sector-specific shocks. This compression is valuable because it reduces 11 correlated series to a few orthogonal signals.

Regime classification. The HMM cleanly identifies stress periods. The high-volatility regime aligns with major drawdowns and shows the largest negative mean return and highest variance. Transition probabilities indicate that bear regimes are persistent once entered.

Predictability. Lagged PCA factors hold meaningful information about regime transitions. On the out-of-sample period, logistic regression yields an AUC of 0.50, indicating real predictive signal. PC1 and PC2 coefficients are economically intuitive: large negative market shocks and rotation toward defensives increase the probability of entering the high-vol regime.

Backtest performance. The regime-aware strategy reduces exposure during high-volatility periods, cutting drawdowns and volatility:

Strategy	AnnRet	AnnVol	Sharpe	MaxDD
Regime-aware	0.199	0.124	1.284	-0.102
SPY Buy/Hold	0.119	0.198	0.399	-0.357

Even when returns are similar, the Sharpe and drawdown improvements are material. The strategy's smoother equity curve (Figure 1, right) reflects its ability to avoid high-volatility crashes.

4. Conclusion

This project shows that a small number of PCA-derived sector factors contain predictive information about market volatility regimes. When combined with an HMM and a simple allocation rule, these factors produce lower volatility, smaller drawdowns, and improved risk-adjusted performance relative to buy-and-hold SPY.

The framework is transparent, computationally light, and easily extendable. Natural next steps include rolling PCA, adding macro predictors (rates, credit spreads), or replacing logistic regression with nonlinear models to capture richer dynamics.